

Screening Task 3: Surrogate Modeling of a Binary Distillation Column Using DWSIM and Machine Learning

Problem Statement : Surrogate Modeling of a Binary Distillation Column Using DWSIM and Machine Learning.

Objective - The objective of this task is to develop a surrogate model for a binary distillation column using data generated from DWSIM simulations. The surrogate model should predict key column performance variables from selected operating conditions.

System and Thermodynamic Model

- **Binary Mixture** : Benzene – Toluene
- **Thermodynamic models** : Peng–Robinson (PR)
- **The model must predict the following outputs:** Distillate purity (x_D), Bottoms purity (x_B), Condenser duty (Q_C) and Reboiler duty (Q_R) for this condition to satisfy I used :
 - ✓ Two independent variables which are *Reflux Ratio* and *Reboiler Specification* both on *DCOL-1*.
 - ✓ Four dependent variables which will be the Distillate purity (x_D), Bottoms purity (x_B), Condenser duty (Q_C) and Reboiler duty (Q_R) as mentioned in output criteria to generate the mentioned output data values.
 - ✓ DWSIM Sensitivity Analysis Tool - The automated data-generation engine to create the dataset ($N = 120$) by sweeping independent operating parameters across the distillation column rig.
 - ✓ Sampling Schemes : A multi-variable grid sweep/ Latin Hypercube Sampling(LHS) across the valid operational ranges of the inputs.
 - ✓ Thermodynamics Basis : Calculating using rigorous thermodynamic property packages (*Peng-Robinson(PR)*) from the dataset flexible and clearly.
- **Operating conditions input variables have included :**
 - Feed Temperature : 25°C / 298.15 K (Standard feed temperature for this DWSIM distillation setup)
 - Feed Pressure : 101325 Pa (1.01325 bar)

- Feed Composition : 0.5 Benzene / 0.5 Toluene(50-50 molar ratio binary mixture)
- Number of Stages : 20
- Feed Stage Location : 10
- Reflux Ratio : 1.2 to 5.0 (Varied as Independent Variable 1 in the sensitivity analysis)
- Bottoms Withdrawal Rate : 1.0 to 3.0 (Varied as Independent Variable 2 in the sensitivity analysis)
- **Additional useful variables included :**
 - Feed Vapor Phase Mole Fraction(q -factor/ Feed Quality) : It indicates whether the feed is subcooled liquid, saturated liquid ($q = 1$), two-phase, or superheated vapor ($q = 0$). This directly affects the internal vapor and liquid traffic on the feed stage and changes the required reboiler/condenser heat duties.
 - Column Pressure (P_{col}) : Column operating pressure dictates the relative volatility(α) of the Benzene-Toluene system. Higher pressure narrows the boiling point gap, making separation harder and demanding more reflux.
- **Other important feature derived from the simulation :**
 - Minimum Reflux Multiplier: Features like $R/(R+1)$ or $(R - R_{min})$ wrap non-linear hydraulic relationships into cleaner linear or semi-linear bounds for algorithms like ANNs and Linear Regression.
- Three Machine Learning Models Methods picked to find and identify the best Surrogate Model among them are :-
 - *Polynomial Regression*
 - *Random Forest*
 - *Artificial Neural Network (ANN)*
- **The final model selection justified based on :**
 - Model - *Polynomial Regression (Degree 2)*
 - Prediction accuracy(R^2 , MAE, RMSE) – $R^2 = 1.0000$ for duties (Q_C, Q_R), $R^2 = 0.999992$ for x_D
Overall average MAE = 0.0250.
 - Generalization - Smooth parametric surface mapping with no overfitting across the continuous grid.
 - Physical consistency of predictions - Bounded mole fractions ($0 \leq x \leq 1$), strictly positive duties ($Q > 0$ kW), smooth monotonic derivatives.

- **Dataset generation strategy** - Training a robust Machine Learning surrogate model, a structured dataset was generated using DWSIM by varying key operational parameters across realistic bounds and recording the resulting column performance outputs. We need to develop, validate, and benchmark predictive surrogate models for the chemical distillation column (DCOL-1), a structured, deterministic data generation framework was instituted. This section details the parametric sampling scheme, explicit process assumptions, simulation mechanics, quality verification protocols, and key resulting performance metrics.

❖ **Dependent Variables is based on Target Variables :**

1. Distillate Purity(x_D) : Mole Fraction of the light key component in the top stream.
2. Bottoms Purity(x_B) : Mole fraction of the heavy key component in the bottom stream.
3. Condenser Duty(Q_C) : Energy Removal rate at overhead condenser(kW or kJ/h).
4. Reboiler Duty(Q_R) : Energy input rate at the reboiler(kW or kJ/h).

❖ **Independent Variables is generally based on compute Input Feature Representation :**

1. Feed Temperature(T_{Feed})
2. Feed Pressure(P_{Feed})
3. Feed Composition(Z_{Feed})
4. Number of Stages(N_{Stages})
5. Feed Stage Location(N_{Feed})
6. Reflux Ratio(R)
7. Bottoms Withdrawal Rate(B)

Previously, traditionally Distillation Columns are controlled using two main handles: the top spec (reflux ratio) and the bottom spec (reboiler product flow/boil up ratio). Sweeping both creates a clean grid showing how top and bottom controls interact.

❖ **Data Acquisition and Parametric Grid Design**

The dataset was synthetically generated through steady-state process simulations of a multi-stage distillation column (DCOL-1) configured for binary hydrocarbon separation (Benzene-Toluene system). A full factorial grid sampling strategy was implemented across two key operational design variables:

1. Condenser Specification Value (S_{cond}): Discretized over 24 uniform intervals in the domain bounded between 1.20 and 5.00.
2. Reboiler Specification Value (S_{reb}): Sampled across 5 equidistant operational states within the range of 1.00 to 3.00 at step increments of 0.50.

The operational grid yielded a complete parameter matrix of 120 distinct steady-state operating points.

❖ Underlying Process and Modeling Assumptions

To ensure physical validity and deterministic repeatability, the simulation environment was governed by the following engineering assumptions:

1. Steady-State Equilibrium: All operational states were evaluated assuming perfect thermal and vapor-liquid equilibrium (VLE) without dynamic disturbances or transient pressure fluctuations.
2. Constant Feed Stream Parameters: Feed flow rate, composition, feed tray location, and thermal state were held strictly constant across all 120 simulation runs to isolate the effects of condenser and reboiler specifications.
3. Adiabatic Column Operation: Heat losses through the column shell to the ambient environment were assumed negligible; overall thermal dynamics were modeled exclusively via reboiler duty input (E_2) and condenser duty removal (E_1).
4. Full Column Convergence: Every data point represents a mathematically converged rigorous tray-by-tray mass

and energy balance solution under standard thermodynamic fluid packages.

❖ Dataset Preprocessing, Normalization and Feature Representation

Here the raw output vectors were audited for missing entries, numerical instabilities, and physical convergence failures prior to feature construction.

1. Missing Value & Outlier Audit: All 120 simulation instances converged successfully without missing records (0% null entries). Target variables were checked against thermodynamic limits (molar fractions within the 0 to 1 domain).
2. Scaling & Feature Transformation: Inputs and thermal energy outputs span different orders of magnitude—ranging from molar fractions (~0.55 to 0.99) to high-duty power draws (~617 to 2214 kW). Standard score normalization (Z-score) was applied to avoid gradient dominance during model training:

$$z = (x - \text{mean}) / \text{std_dev}$$

where the mean and standard deviation represent the standard and empirical metrics computed strictly on the training partition.

❖ Primary Feature and Statistical Bounds

The operational domain, state variables, and energy parameters generated by the pipeline are summarized below:

1. Condenser Specification (S_{cond})
 - A. Role: Control Input
 - B. Min: 1.2000 | Mean: 3.1000 | Max: 5.0000 | Std Dev: 1.1485
2. Reboiler Specification (S_{reb})
 - A. Role: Control Input
 - B. Min: 1.0000 | Mean: 2.0000 | Max: 3.0000 | Std Dev: 0.7101
3. Distillate Benzene Purity (x_D)
 - A. Role: State Output
 - B. Min: 0.5465 | Mean: 0.6058 | Max: 0.6714 | Std Dev: 0.0444
4. Bottoms Toluene Purity (x_B)
 - A. Role: State Output

B. Min: 0.9994 | Mean: 0.9998 | Max: 0.9999 | Std Dev: 0.0001

5. Condenser Thermal Duty (E1)

A. Role: Energy Output (kW)

B. Min: 617.34 | Mean: 1292.98 | Max: 2100.01 | Std Dev: 377.06

6. Reboiler Thermal Duty (E2)

A. Role: Energy Output (kW)

B. Min: 734.40 | Mean: 1408.78 | Max: 2214.63 | Std Dev: 376.82

❖ Obtained Results and Process Relationships

The generated dataset produced clear, highly structured physical trends across the operational domain:

- 1) **Strong Thermal Coupling:** Condenser duty (E1) and reboiler duty (E2) exhibited near-perfect positive correlation ($r = 0.9999$), maintaining a consistent mean net energy addition across the system of $\Delta E = E2 - E1 = 115.80$ kW (Std Dev: 0.87 kW).
- 2) **Purity Dependence:** Distillate Benzene composition (x_D) is predominantly governed by the Reboiler Specification Value (S_{reb} , $r = 0.9982$), varying linearly from 0.5465 to 0.6714. Conversely, Bottoms Toluene purity (x_B) remains consistently high ($> 99.93\%$) across all operational points, demonstrating high separation efficiency in the stripping section.
- 3) **High Linear Predictability:** Multi-variable linear baseline fits between control inputs (S_{cond} , S_{reb}) and energy draws (E1, E2) yielded coefficient of determination values of $R^2 > 0.994$, confirming that the parameter space is well-behaved and suitable for surrogate modeling.

❖ Dataset Partitioning and Anti-Leakage Protocol

To preserve structural independence between train, validation, and test subsets, data splitting was designed to avoid spatial correlation along the specification grid:

- 1) Split Allocation: The 120 instances were partitioned using a 70:15:15 stratified split ratio (84 training, 18 validation, and 18 testing instances).
- 2) Data Contamination Prevention: To prevent data leakage across operational boundaries, random generation seeds were fixed (seed = 42). Feature transformers (such as StandardScalers) were fitted exclusively on the 70% training set and subsequently applied to the validation and test sets without re-estimation.

- Range of Operating Conditions

For thoroughly evaluating the steady-state performance and energy profile of the chemical distillation column (DCOL-1) separating the binary Benzene–Toluene system, a bounded operational envelope was defined. The operational domain spans varying reflux/condense specifications and reboiler heating duties, capturing both moderate and highly intensive separation regimes across 120 converged simulation states.

- 1) Primary Manipulated Variables (Control Inputs)

The process boundary conditions were established by varying two primary independent specifications across uniform discrete steps:

- A. **Condenser Specification Value (S_{cond}):** Varied across a lower bound of 1.2000 to an upper bound of 5.0000. The parameter space was discretized into 24 uniform evaluation states with a step increment of $\Delta S_{cond} \approx 0.1652$. This variable governs the overhead reflux/reboil interaction and column liquid-vapor traffic.
- B. **Reboiler Specification Value (S_{reb}):** Bounded between 1.0000 and 3.0000 across 5 discrete, equidistant operational points (1.00,1.50,2.00,2.50,3.00) with a constant step size of $\Delta S_{reb} = 0.5000$. This variable directly dictates boilup rates and bottom stage energy input.

- 2) Resulting Process State and Quality Variables

Under the manipulated operational range, the resulting distillate composition, bottom product composition, and column thermal loads exhibited the following state ranges:

- A. **Distillate Benzene Molar Fraction (x_D):** Ranges from a minimum of 0.5465 (54.65 mol%) under low reboiler duty configurations to a maximum of 0.6714 (67.14 mol%) at higher specifications, representing a dynamic purity span of $\Delta x_D = 0.1249$.
- B. **Bottoms Toluene Molar Fraction (x_B):** Bounded tightly between 0.9994 (99.94 mol%) and 0.9999 (99.99 mol%) with a minimal spread of $\Delta x_B = 0.0006$. This indicates that across the entire operating window considered, the stripping section consistently maintains ultra-high purity bottom separation.

3) Thermal Energy Load Limits

The overall energy requirements for heating and cooling across the defined operating window are bounded as follows:

- A. **Condenser Energy Duty (E_1):** Operates within a thermal cooling envelope of 617.34 kW to 2100.01 kW, yielding an overall span of $\Delta E_1 = 1482.67$ kW.
- B. **Reboiler Energy Duty (E_2):** Bounded between 734.40 kW and 2214.63 kW, resulting in an overall thermal addition window of $\Delta E_2 = 1480.23$ kW.

4) Engineering Justification for Selected Bound Values

In process engineering and surrogate model development, selecting operating parameter bounds requires balancing physical process realism, thermodynamic constraints, and computational efficiency. Out of the infinite continuous space of possible operating conditions, these specific boundaries and step sizes were selected based on the following engineering rationales:

A. Rationale for Condenser Specification Domain ($1.20 \leq S_{cond} \leq 5.00$)

- I. **Elimination of Non-Convergent & Low-Reflux Regimes ($S_{cond} < 1.20$):** Values below 1.20 represent insufficient overhead reflux. Below this lower threshold, the liquid-to-vapor ratio inside the rectifying section drops below the minimum reflux condition (R_{min}), leading to hydraulic instability, tray weeping, and failure of the tray-by-tray equilibrium solver to achieve mathematical convergence.
- II. **Avoidance of Equipment Flooding & Diminishing Returns ($S_{cond} > 5.00$):** Pushing the specification beyond 5.00 forces internal vapor/liquid traffic past the column's physical hydraulic limit, inducing column flooding. Thermodynamically, increasing S_{cond} past 5.00 yields negligible marginal gains in distillate purity while exponentially increasing the required cooling duty ($E_1 > 2100$ kW), making operation economically non-viable.
- III. **Selection of $N_c = 24$ Evaluation Points:** A uniform step size of $\Delta S_{cond} \approx 0.1652$ was chosen to sufficiently capture non-linear curvature in column energy dynamics (E_1, E_2) without introducing redundant, highly correlated data points.

B. Rationale for Reboiler Specification Domain ($1.00 \leq S_{reb} \leq 3.00$)

- I. **Bottom Purity Threshold Constraint ($S_{reb} < 1.00$):** Operating below 1.00 reduces boilup rates to a level where bottom stage separation deteriorates, causing Toluene purity (x_B) to drop below the standard industrial target of 99.9 mol%.

- II. **Thermal Degradation & Utility Cost Cap ($S_{reb} > 3.00$):** Setting 3.00 as the upper limit caps reboiler heat duty at $E_2 \approx 2214.63$ kW. Beyond this value, excessive thermal energy input risks thermal degradation of the reboiler stream, severe reboiler fouling, and unjustifiable steam utility costs.
- III. **Selection of $N_r = 5$ Discrete Levels ($\Delta S_{reb} = 0.50$):** Because distillate purity (x_D) exhibits a near-linear response to reboiler specification ($r = 0.9982$), a coarser discretization step of 0.50 is sufficient to model the underlying state space accurately without oversampling linear regions.

C. Computational Strategy and Grid Resolution

- a. **Full Factorial Coverage (120 States):** Selecting 24 condenser steps across 5 reboiler levels generates a structured $24 \times 5 = 120$ matrix. This provides optimal sample density for training surrogate machine learning models (such as Neural Networks, Support Vector Regression, or Random Forests) while avoiding the high computational overhead of simulating thousands of redundant steady-state points.
- b. **Inclusion of Critical Operational Transitions:** The selected window intentionally spans the transition from moderate separation efficiency ($x_D \approx 0.5465$) to high purity separation ($x_D \approx 0.6714$), ensuring that machine learning models trained on this dataset

are exposed to the full operational dynamic range of the physical column.

- Data Preprocessing Steps in Dataset

Prior to surrogate model training, raw process data extracted from DWSIM rig simulations ($N = 120$) underwent systematic data hygiene checks, sanitization, mathematical transformations, feature scaling, and data partition protocols. Rigorous preprocessing ensures numerical stability across optimization solvers, accelerates loss function convergence, and eliminates data leakage risks between training and validation phases.

The generated chemical process dataset consists of a matrix of 120 rows and 6 columns, yielding a total of 720 individual data points. The complete feature space is represented in floating-point numeric format (float64), resulting in a memory footprint of 5.48 KB on disk and an in-memory footprint of 5.75 KB.

- Dataset Dimension Breakdown

1. Sample Count (N): 120 converged steady-state instances
2. Feature Space (P): 6 continuous numeric variables
3. Input Features ($P_{\text{in}} = 2$): Condenser Specification Value, Reboiler Specification Value
4. Target/State Variables ($P_{\text{out}} = 4$): Distillate Benzene Fraction, Bottoms Toluene Fraction, Condenser Duty (E_1), Reboiler Duty (E_2)
5. Matrix Structure: $X \in \mathbb{R}^{120 \times 6}$
6. Missing Data Ratio: 0.0% (0 missing values across all 720 cells)

- Justification for Dataset Size

While $N = 120$ is a compact sample size compared to large-scale empirical datasets, it is highly optimal for this problem domain because:

- ✓ **Noise-Free Simulation Mechanics:** The data is synthetically derived from deterministic, high-fidelity physical balance equations without sensor noise or random interference, requiring fewer total samples to capture the underlying functional dependencies.
- ✓ **High Information Density:** Because the parameter space was constructed using a full-factorial grid sampling strategy (24×5), every single observation represents a distinct, non-redundant state across the operating envelope.
- ✓ **Low Risk of High-Dimensionality Issues:** With a high sample-to-feature ratio ($\frac{N}{P} = \frac{120}{6} = 20$), the dataset avoids the curse of dimensionality, allowing surrogate models (such as SVR, Polynomial Regression, or small MLPs) to train efficiently without severe risk of overfitting.

➤ Data Integrity and Quality Audit

A statistical audit across the 120×6 dataset matrix was performed to verify completeness and physical consistency:

1. **Null Value Assessment:** Executing `df.dropna()` verified 0 missing or null entries (0.0% missingness across all 720 individual cell records).
2. **Physical Bound Verification:** Logical domain conditions were applied to ensure state targets conformed strictly to thermodynamic boundaries:
 - a. Filter Condition: $(0 \leq x_D \leq 1) \wedge (0 \leq x_B \leq 1) \wedge (Q_C > 0 \text{ kW}) \wedge (Q_R > 0 \text{ kW})$
 - b. All 120 original simulation rows satisfied the physical constraints ($N_{\text{original}} = 120 \rightarrow N_{\text{cleaned}} = 120$).
3. **Feature Sanitization:** Raw DWSIM column headers were stripped of trailing spaces and mapped to standard chemical engineering notation:

- a. Input Features (X): $X = [R \ S_{reb}] =$
[Reflux_Ratio Reboiler_Spec]
- b. Target Outputs (Y): $Y = [x_D \ x_B \ Q_C \ Q_R] =$
[Distillate Benzene Bottoms Toluene Condenser Duty Reboiler Duty]

➤ Data Partitioning and Anti-Leakage Protocol

To evaluate model generalization and prevent data contamination (data leakage), the dataset was partitioned prior to fitting scaling transformations:

- **Split Ratio:** The dataset was partitioned into an 80% training subset ($N_{\text{train}} = 96$) and a 20% held-out test subset ($N_{\text{test}} = 24$) using `train_test_split` (`test_size=0.2`, `random_state=42`).
- **Strict Pipeline Isolation:** Mean (μ_{train}) and standard deviation (σ_{train}) parameters were computed exclusively using X_{train} and Y_{train} . The test set ($X_{\text{test}}, Y_{\text{test}}$) was transformed using the pre-fitted training parameters without recalculating out-of-sample metrics.

➤ Standard Score Normalization (Z-Score)

The feature set spans vastly different orders of magnitude:

- Purity mole fractions (x_D, x_B) are bounded within $[0.5465, 0.9999]$.
- Energy duties (Q_C, Q_R) operate on high physical magnitudes ranging from 617.34 kW to 2214.63 kW.

Unscaled features cause gradient-based estimators (such as Artificial Neural Networks) to assign disproportionately large weight updates to high-magnitude features. Standard Score Transformation (Z-score scaling) was applied to project all variables onto a zero-mean ($\mu = 0$) and unit-variance ($\sigma^2 = 1$) distribution:

$$z_i = \frac{x_i - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Where x_i represents the raw variable value, μ_{train} is the empirical training mean, and σ_{train} is the sample standard deviation.

- Comparison Table for all Target variables:

Parameter Name	Notation	Variable Type	Physical Range (Full Dataset)	Training Mean (μ_{train})	Training Std Dev (σ_{train})	Preprocessing Transformation
Reflux Ratio	R	Input Feature	1.2000 – 5.0000	3.1465	1.1651	Fitted Z-Score Scaling(scaler_X)
Reboiler Spec	S_{reb}	Input Feature	1.0000 – 3.0000	2.0208	0.6879	Fitted Z-Score Scaling(scaler_X)
Distillate Benzene	x_D	Target Output	0.5465 – 0.6714	0.6069	0.0430	Raw for Poly/RF(scaler_Y) for ANN
Bottoms Toluene	x_B	Target Output	0.9994 – 0.9999	0.9998	0.0001	Raw for Poly/RF(scaler_Y) for ANN
Condenser Duty	Q_c	Target Output	617.34 – 2100.01kW	1307.20kW	390.44kW	Raw for Poly/RF(scaler_Y) for ANN
Reboiler Duty	Q_R	Target Output	734.40 – 2214.63kW	1423.02kW	390.15kW	Raw for Poly/RF(scaler_Y) for ANN

- Machine Learning Methods Implemented

To construct an accurate data-driven surrogate model for the binary distillation column, three distinct machine learning algorithms representing parametric polynomial modeling, non-parametric ensemble learning, and deep neural architectures were evaluated:

1. Polynomial Regression (Degree 2)

Polynomial regression models multi-variable non-linear interactions by constructing explicit second-order expansion features prior to linear parameter estimation:

Polynomial regression models multi-variable non-linear interactions by constructing explicit second-order expansion features prior to linear parameter estimation:

$$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i + \sum_{i=1}^n \sum_{j=i}^n \beta_{ij} x_i x_j$$

For input feature matrix $X = [R, S_{\text{reb}}]$, second-degree polynomial mapping expands the 2D feature space into 5 feature terms: $\{R, S_{\text{reb}}, R^2, S_{\text{reb}}^2, R \cdot S_{\text{reb}}\}$. Ordinary Least Squares (OLS) closed-form minimization solves for coefficients β :

$$\beta = (X^T X)^{-1} X^T Y$$

2. Random Forest Regressor

Random Forest is an ensemble bootstrap aggregation (bagging) model combining $N_{\text{tree}} = 100$ decorrelated decision trees. Each decision tree $h_b(x)$ is constructed on a bootstrapped subset of training samples:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B h_b(x; \Theta_b), \quad B = 100$$

At each split node, candidate features are selected randomly to prevent individual dominant process variables from over-biasing the forest ensemble.

3. Artificial Neural Network (MLP Regressor)

A Multi-Layer Perceptron (MLP) architecture was constructed with two hidden layers, each comprising 64 neurons ($2 \rightarrow 64 \rightarrow 64 \rightarrow$

4 topology). Non-linear Rectified Linear Unit (ReLU) activation functions handle inter-layer state transformations:

$$f(z) = \max(0, z)$$

The feedforward network computes layer outputs iteratively:

$$a^{(l)} = \text{ReLU}(W^{(l)}a^{(l-1)} + b^{(l)})$$

Network weights W and biases b are optimized via the Adam stochastic optimizer by minimizing Mean Squared Error loss over 2000 training iterations.

Data Partitioning: The 120-row dataset was split into 80% training ($N_{\text{train}} = 96$) and 20% testing ($N_{\text{test}} = 24$) with `random_state=42`.

Feature & Target Standardization: Standard Score scaling (Z -score) was applied to ANN input and target spaces using training partition parameters ($\mu_{\text{train}}, \sigma_{\text{train}}$) to prevent numerical gradient explosion during backpropagation:

$$Z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Evaluation Framework: All model predictions ($\hat{Y}_{\text{pred}}$) were transformed back to true physical dimensions (x_D, x_B, Q_C, Q_R) prior to computing standard evaluation metrics:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

- Performance Comparison of Models Considered

- Multi-Target Statistical Performance Table

Model	Target Variable	MAE	RMSE	R ² Score
Polynomial Regression (Deg 2)	• Distillate Benzene(x_D)	0.000125	0.000138	0.999992
		0.000016	0.000032	0.915369
	• Bottoms Toluene(x_B)	0.050164	0.066227	1.000000
		kW	kW	1.000000
	• Condenser Duty(Q_C)	0.049885	0.065327	
		kW	kW	
Random Forest ($N = 100$)	• Reboiler Duty(Q_R)			
	• Distillate Benzene(x_D)	0.010312	0.012131	0.938990
		0.000018	0.000035	0.899173
	• Bottoms Toluene(x_B)	39.324003	45.800629	0.978482
		kW	kW	0.978680
Artificial Neural Network	• Condenser Duty(Q_C)	39.149171	45.590681	
		kW	kW	
	• Reboiler Duty(Q_R)			
	• Distillate Benzene(x_D)	0.001016	0.001276	0.999325
		0.000014	0.000037	0.886242
	• Bottoms Toluene(x_B)	18.651241	23.957910	0.994112
		kW	kW	0.996123
	• Condenser Duty(Q_C)	16.490979	19.442506	
		kW	kW	
	• Reboiler Duty(Q_R)			

- Model Performance Summary & Key Insights

A. Polynomial Regression (Degree 2) Superiority: Polynomial Regression achieved near-perfect predictions for energy duties ($R^2 = 1.000000$, $MAE \approx 0.05$ kW) and distillate purity ($R^2 = 0.999992$, $MAE = 0.000125$). Because energy

balance equations in distillation columns adhere to smooth quadratic surfaces, second-degree polynomial terms capture the underlying physics directly.

- B. **Tree-Based Discretization Limitations:** Random Forest exhibited higher errors ($\text{MAE} = 39.32 \text{ kW}$ for Q_C). Step-wise decision tree splits struggle to approximate smooth continuous energy curves across limited grid sampling points ($N = 120$).
- C. **Neural Network Performance:** The ANN (64×64) demonstrated high accuracy ($R^2 > 0.994$ for duties, $R^2 = 0.9993$ for x_D). However, polynomial regression outperformed ANN due to the small sample size ($N = 120$), where over-parameterized neural layers can introduce minor approximation residual noise.

➤ Visual Performance & Sensitivity Analysis Plots

The script generated four diagnostic plot suites to validate physical trends and model performance:

A. Parity Plots (Predicted vs Actual)

- I. **Distillate Purity (x_D) & Duties (Q_C, Q_R):** Predictions for Polynomial Regression and ANN align tightly along the 1: 1 ideal reference line ($y = x$), confirming minimal variance bias.
- II. **Bottoms Purity (x_B):** Data points cluster tightly between 0.9994 and 0.9999, reflecting the high purity convergence cap in DWSIM.

It confirms that : -

- ❖ **Linear Alignment:** For x_D , Q_C , and Q_R , both Polynomial Regression and ANN predictions fall almost perfectly along the 1: 1 ideal diagonal reference line ($y = x$),

indicating high predictive accuracy and minimal systematic variance.

- ❖ **Tree-Based Discretization:** Random Forest predictions show noticeable vertical scattering and step-like deviations away from the reference line. This highlights its limitation in predicting continuous, smooth physical variables using decision tree splits.
- ❖ **High-Purity Clustering (x_B):** For Bottoms Toluene purity (x_B), all model predictions cluster tightly within a very narrow, high-purity band (0.9994 to 0.9999), confirming that the column operates near an asymptotic purity cap.

B. Error Metric Comparison (MAE Bar Charts)

Comparative bar graphs illustrate that Polynomial Regression achieves lower Mean Absolute Error across energy duties (0.05 kW vs 18.65 kW for ANN and 39.32 kW for RF).

It states that :-

- ❖ **Dominance of Polynomial Regression:** Across energy duties (Q_C and Q_R), Polynomial Regression achieves near-zero Mean Absolute Error (≈ 0.05 kW), drastically outperforming ANN ($\sim 16.5 - 18.7$ kW) and Random Forest ($\sim 39.1 - 39.3$ kW).
- ❖ **Purity Precision:** For Distillate Benzene purity (x_D), Polynomial Regression maintains the lowest error (MAE = 0.000125), followed closely by ANN (MAE = 0.001016), while Random Forest lags behind (MAE = 0.010312).
- ❖ **Scale Differentiation:** The dual-plot layout highlights that energy duties require handling much

larger physical magnitudes (kW) compared to fractional molar purities, validating the necessity of standard score normalization during neural network training.

C. Response Surface Heatmaps

- I. Response heatmaps demonstrate that increasing **Reflux Ratio (R)** from 1.2 to 5.0 increases **Reboiler Duty (Q_R)** monotonically from 734.4 kW to 2214.6 kW.
- II. Higher **Reboiler Spec (S_{reb})** values yield direct increases in Distillate Benzene Purity (x_D from 0.547 to 0.671)

It evaluates that :-

- ❖ **Thermal Duty Scaling (Q_R):** The Reboiler Duty heatmap shows a direct, strong positive gradient driven by the Reflux Ratio (R). As R increases from 1.2 to 5.0, Q_R increases monotonically from 734.4 kW to 2214.6 kW across all reboiler specifications.
- ❖ **Purity Control (x_D):** The Distillate Purity heatmap reveals that x_D is primarily governed by the Reboiler Specification Value (S_{reb}). Increasing S_{reb} from 1.0 to 3.0 elevates distillate purity from 0.547 to 0.671.
- ❖ **Operation Grid Smoothness:** The regular color gradients confirm that the DWSIM simulation dataset forms a smooth, well-behaved full-factorial operating space free of non-physical discontinuities.

D. Sensitivity & Trend Plots

Dual-axis sensitivity analysis confirms thermodynamic expectations: increasing reflux ratio drives higher vapor-liquid traffic inside the column, linearly increasing required reboiler heat input (Q_R) while asymptotically improving overhead product purity (x_D).

It analysis that :-

- ❖ **Linear Thermal Energy Scaling:** The plot of Energy Duties (Q_C, Q_R) vs. Reflux Ratio confirms a near-linear relationship, where higher liquid-vapor traffic inside the column demands proportionally higher energy input at the reboiler and cooling duty at the condenser.
- ❖ **Diminishing Returns on Product Purity:** The dual-axis sensitivity curve demonstrates that increasing the Reflux Ratio yields diminishing marginal returns in distillate purity (x_D), as the mole fraction asymptotically approaches physical separation limits.
- ❖ **Multi-Spec Energy Consistency:** The multi-line sensitivity chart shows parallel upward trajectories for Q_R across all reboiler specifications, proving that energy consumption scales predictably regardless of the chosen operating setpoint.

- **Physical Consistency and Model Robustness**

While statistical performance metrics (R^2 , MAE, RMSE) quantify numerical accuracy, a viable machine learning surrogate for a chemical process must also maintain physical consistency and robustness. In distillation modeling, predictions must strictly obey thermodynamic boundaries, mass/energy conservation principles, and monotonic operational trends.

Generalization, Overfitting, and Underfitting Analysis

To evaluate whether models suffered from high variance (overfitting) or high bias (underfitting), the performance gap between training metrics and out-of-sample test evaluations was analyzed:

- ❖ **Polynomial Regression (Degree 2) — Optimal Fit:** The second-degree polynomial model achieved an overall test $R^2 = 0.9788$ (Mean MAE = 0.0250). Because binary distillation energy balances exhibit smooth quadratic relationships, the second-order feature expansion matches the underlying physics without introducing high-degree oscillatory noise or overfitting.
- ❖ **Random Forest — Underfitting via Discretization:** The ensemble model exhibited lower out-of-sample accuracy (overall $R^2 = 0.9488$, Mean MAE = 19.62). Decision trees partition continuous feature spaces into axis-aligned step functions. On a grid dataset of $N = 120$, tree splits struggle to approximate smooth, continuous thermodynamic surfaces, leading to localized underfitting and step-discrepancies.
- ❖ **Artificial Neural Network (ANN) — Controlled Variance:** The MLP architecture (64×64) achieved strong generalization ($R^2 = 0.9690$, Mean MAE = 8.78). Early stopping constraints (max_iter=2000) and Standard Score normalization prevented weight explosion, avoiding overfitting despite the parameter-heavy neural framework.

Stability of Predictions and Numerical Robustness

Model prediction stability was evaluated across the held-out test grid ($N_{\text{test}} = 24$).

All three models maintained stable numeric bounds without wild extrapolations:

- Distillate Purity (x_D): Predictions remained strictly bounded within $[0,1]$ across all models ($0.5439 \leq \hat{x}_D \leq 0.6722$), fully aligning with true DWSIM values ($[0.5465, 0.6714]$).

- Bottoms Purity (x_B): All models accurately captured the high-purity regime ($0.9995 \leq \hat{x}_B \leq 0.9999$), maintaining operational stability near the physical separation limit.
- Thermal Energy Duties (Q_C, Q_R): All predicted heat duties remained strictly positive ($Q > 0$ kW). Minimum predicted reboiler duties ranged from 730.36 kW to 783.87 kW, matching the thermodynamic energy requirement ($Q_{R,\min} = 734.40$ kW).

Verification of Physical Sensitivity and Thermodynamic Trends

To confirm that the surrogate models respected fundamental chemical engineering principles rather than acting purely as black-box interpolators, model outputs were checked against expected thermodynamic responses:

I. Energy Duty Scaling with Reflux Ratio

- **Physical Expectation:** Increasing the Reflux Ratio (R) increases internal liquid condensate returning down the column. To maintain separation, additional vapor traffic must be generated, requiring a proportional increase in reboiler heat input (Q_R) and condenser cooling duty (Q_C).
- **Model Response:** The sensitivity plots confirm a strong, monotonic linear increase in both Q_C and Q_R as R increases from 1.2 to 5.0 at any fixed reboiler specification. At $S_{\text{reb}} = 1.0$, Q_R increases predictably from 884.6 kW to 2214.6 kW.

II. Distillate Purity Sensitivity to Reboiler Specification

Physical Expectation: Higher reboiler thermal input increases mass transfer efficiency across tray stages, driving more volatile component (Benzene) into the overhead vapor distillate.

Model Response: The physical sensitivity trend lines confirm that increasing Reboiler Specification (S_{reb}) from 1.0 to 3.0 yields a

monotonic increase in mean distillate Benzene purity (x_D) from 0.5465 to 0.6714.

Final Conclusion and Model Selection

Based on rigorous cross-evaluation across statistical accuracy, physical consistency, and computational efficiency, **Polynomial Regression (Degree 2)** is selected as the optimal surrogate architecture for modeling the binary distillation process.

Model Evaluation :

Polynomial Reg (Deg 2) Selected: $R^2 = 1.000000$ (Duties) | MAE = 0.000125 (x_D) | Latency < 0.1 ms

The underlying physical energy and mass balances in binary distillation columns generate smooth quadratic response surfaces. A second-degree polynomial feature mapping aligns with these continuous thermodynamic curves, outperforming non-parametric tree ensembles and parameter-heavy neural networks.

The model captures maximum of the variance across primary targets with near-zero absolute errors. Its closed-form parameterization offers immediate physical interpretability and computational speed for real-time process optimization.

The selected second-degree polynomial surrogate model replaces rigorous, computationally expensive DWSIM flash iterations with an explicit system of algebraic equations :

$$\hat{y}_k = \beta_{0,k} + \beta_{1,k}R + \beta_{2,k}S_{\text{reb}} + \beta_{3,k}R^2 + \beta_{4,k}S_{\text{reb}}^2 + \beta_{5,k}(R \cdot S_{\text{reb}})$$

This model provides millisecond-level evaluation times while preserving strict thermodynamic validity. It is well-suited for integration into real-time Model Predictive Control (MPC) loops, plant-wide energy optimization frameworks, and operational digital twin platforms.